

CS 771A: Intro to Machine Learning, IIT Kanpur			Midsem Exam (22 Feb 2026)	
Name				40 marks
Roll No		Dept.		Page 1 of 4

Instructions:

1. This question paper contains 2 pages (4 sides of paper). Please verify.
2. Write your name, roll number, department in **block letters** with **ink** on **each page**.
3. Write your final answers neatly **with a blue/black pen**. Pencil marks may get smudged.
4. Don't overwrite/scratch answers especially in MCQ – ambiguous cases will get 0 marks.



Q1 (Mending Melbo's Mistake) In Quiz 1, Melbo made a mistake while trying to simplify the SVM by maximizing the mean margin. The corrected optimization problem is given below. Assume that the data is such that perfect classifiers exist and $\sum_{i \in [N]} y^i \cdot \mathbf{x}^i \neq \mathbf{0}$. **((1+2) + 3 x 2 = 9 marks)**

$$\min_{\mathbf{w}} \left\{ -\sum_{i \in [N]} y^i \cdot \mathbf{w}^\top \mathbf{x}^i \right\} \quad \text{s.t.} \quad y^i \cdot (\mathbf{w}^\top \mathbf{x}^i) \geq 0 \quad \forall i \in [n] \quad \text{and} \quad \|\mathbf{w}\|_2^2 \leq 1$$

Melba thinks the optimum solution $\mathbf{w}^*$ will always satisfy $\|\mathbf{w}^*\|_2 = 1$ even though the constraint asks for $\|\mathbf{w}\|_2^2 \leq 1$. Write **T** or **F**. If T, give proof. If F, give a counter example.

Write down the Lagrangian by introducing dual variables. No derivation needed.

Give an expression for $\mathbf{w}$ in terms of the dual variables and constants e.g. $y^i, \mathbf{x}^i$ etc. No derivation needed.

Simplify the dual problem (eliminate $\mathbf{w}$) – show major steps.

Q2 (Multiclass Least Squares) Multiclassification can be solved as regression problems by assigning an embedding $\mathbf{v}^c \in \mathbb{R}^K$ to each class $c \in [C]$. Embeddings are collected in a matrix $V \in \mathbb{R}^{K \times C}$. For N train points $(\mathbf{x}^i, y^i), i = 1, \dots, N$ with $\mathbf{x}^i \in \mathbb{R}^d, y^i \in [C]$, the embedding assigned to the i^{th} point is $V\mathbf{e}_{y^i}$ where $\mathbf{e}_c \in \mathbb{R}^C$ is the standard basis vector with 1 at the c^{th} coordinate and 0 elsewhere. Melba wants to learn K models $\mathbf{w}^k \in \mathbb{R}^d$, collected as a $K \times d$ matrix $W \in \mathbb{R}^{K \times d}$ by solving $\min_{W, V} \frac{1}{2} \sum_{k \in [K]} \|\mathbf{w}^k\|_2^2 + \frac{1}{2} \sum_{c \in [C]} \|\mathbf{v}^c\|_2^2 + \sum_{i \in [N]} \|W\mathbf{x}^i - V\mathbf{e}_{y^i}\|_2^2$ using alternating optimization.

a. Keeping the embeddings V fixed, find the optimal value of $W \in \mathbb{R}^{K \times d}$. Show steps. (4 marks)

b. Keeping the models W fixed, find the optimal value of $V \in \mathbb{R}^{K \times C}$. Show steps. (4 marks)

c. Melbo notes that Melba's objective doesn't encourage the embeddings $\mathbf{v}^c, c \in [C]$ to be far apart from each other. Melbo wants to use a different regularizer by replacing the term $\frac{1}{2} \sum_{c \in [C]} \ \mathbf{v}^c\ _2^2$. Which of the following regularizers (if added to Melba's objective after removing $\frac{1}{2} \sum_{c \in [C]} \ \mathbf{v}^c\ _2^2$) will encourage the embeddings to be far apart? Fill/shade one or more circles (no derivation needed). Note: $\mathbf{1} \in \mathbb{R}^C$ is the all-ones vector. (4 marks)	$+\frac{1}{2C} \sum_{c \in [C]} \sum_{d \in [C]} \ \mathbf{v}^c - \mathbf{v}^d\ _2^2$	☐
	$-\frac{1}{2C} \sum_{c \in [C]} \sum_{d \in [C]} \ \mathbf{v}^c - \mathbf{v}^d\ _2^2$	☐
	$-\frac{1}{2} \sum_{c \in [C]} \ \mathbf{v}^c\ _2^2 + \frac{1}{2C} \cdot \ V\mathbf{1}\ _2^2$	☐
	$+\frac{1}{2} \sum_{c \in [C]} \ \mathbf{v}^c\ _2^2 + \frac{1}{2C} \cdot \ V\mathbf{1}\ _2^2$	☐
	None of the above	☐

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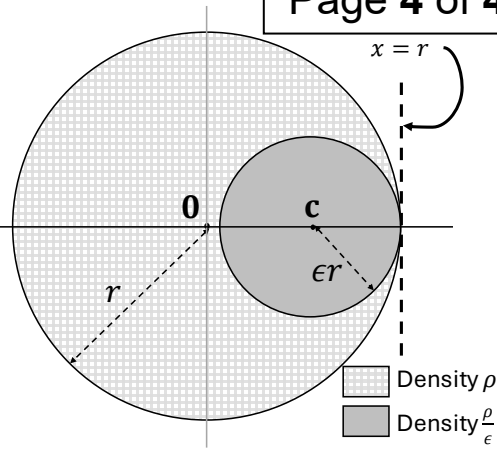
Q3. (True-False) Write **T** or **F** for True/False in the box on the right and a brief justification below.

Note: The expression $p \gg q$ means p is “much larger” than q .

(3 x (1+2) = 9 marks)

1	Using the tanh function $\tau(c) \stackrel{\text{def}}{=} \frac{e^{2c}-1}{e^{2c}+1}$, defining $\mathbb{P}[y^i \mathbf{x}^i, \mathbf{w}] \stackrel{\text{def}}{=} \tau(y^i \cdot \mathbf{w}^\top \mathbf{x}^i)$ is a valid choice of likelihood for a binary classification problem with $\{-1, +1\}$ labels.	
2	For a data point $(\mathbf{x}, y)$, redefining softmax loss using the square function looks like $-\ln\left(\frac{\eta_y}{\sum_{c \in [C]} \eta_c}\right)$ with $\eta_c \stackrel{\text{def}}{=} (\mathbf{w}_c^\top \mathbf{x})^2$. If this loss goes to zero for $(\mathbf{x}, y)$, then it must be the case that $\mathbf{w}_y^\top \mathbf{x} \gg \mathbf{w}_c^\top \mathbf{x}$ for all $c \neq y$. Give a proof if T or a counterexample if F .	
3	A bin with N red, N green and N blue balls has entropy $\log_2 3$. Moreover, no matter how many red/green/blue balls are added to the bin hereon (no other colour is allowed, no ball is taken out), the entropy of the bin can never fall below $\log_2 3$.	

Q4 (Insert Clever Name) Melbruh has a 2D distribution $\mathcal{D}$ with support inside a unit area circle of radius r centered at the origin. $\mathcal{D}$ is described by parameters $\rho \geq 0$ and $\epsilon \in [0,1]$. The density is $\frac{\rho}{\epsilon}$ in the *dense* circle of radius ϵr whose center $\mathbf{c}$ is on the x-axis. The density is ρ in the rest of the support. The two circles are co-tangential with the line $x = r$ as common tangent. Find out valid range of ρ for which $\mathcal{D}$ is a proper dist. with value of $\epsilon \in [0,1]$. Find an expression for $\epsilon \in [0,1]$ in terms of ρ and constants.



For valid combinations of ρ, ϵ , find mean vector $\boldsymbol{\mu} \in \mathbb{R}^2$ of $\mathcal{D}$. *Hint: The mean of the uniform dist. over a circle is its centre.* Give final answers in boxes, derivations later. **(2 + 2 + 3 + 3 = 10 marks)**

$\leq \rho \leq$

$\epsilon =$

$\boldsymbol{\mu} =$

Give brief derivation for valid ranges of ρ , valid values of ϵ for which $\mathcal{D}$ is a proper distribution and its mean $\boldsymbol{\mu}$.